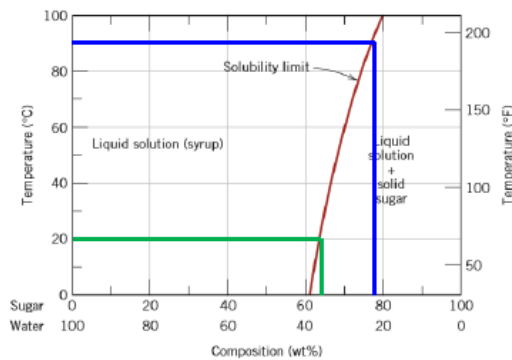
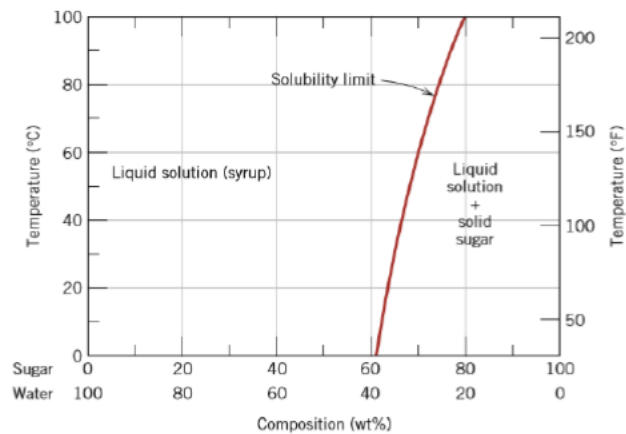


## WS N°2: Phase Diagrams ( Part I )

1. Consider the sugar–water phase diagram in the figure below.
  - a. How much sugar will dissolve in 1500 g water at 90 °C ( 194 ° F ) ?
  - b. If the saturated liquid solution in part ( a ) is cooled to 20 °C ( 68 ° F ), some of the sugar will precipitate out as a solid. What will be the composition of the saturated liquid solution at 20 °C ?
  - c. How much of the solid sugar will come out of solution upon cooling to 20 °C ?



### Solution:

- a) Using the figure above (blue lines):

At 90°C, the maximum concentration of sugar in the sugar-water syrup is 77 wt% of sugar (or 23 wt% of water).

The mass of sugar that will dissolve in 1500 g of water,  $m_{sugar}$ :

$$C_{sugar} = \frac{m_{sugar}}{m_{sugar} + m_{water}}, 0.77 = \frac{m_{sugar}}{m_{sugar} + 1500} \Rightarrow m_{sugar} = 5021.74 \text{ g}$$

Or

$$C_{water} = \frac{m_{water}}{m_{sugar} + m_{water}}, 0.23 = \frac{1500}{m_{sugar} + 1500} \Rightarrow m_{sugar} = 5021.74 \text{ g}$$

b) Using the figure above (green lines):

At 20°C, the composition of the saturated liquid solution is 64 wt% of sugar and 36 wt% of water.

c) The mass of sugar that will dissolve in this saturated solution,  $m'_{sugar}$ :

$$C'_{sugar} = \frac{m'_{sugar}}{m'_{sugar} + m_{water}}, 0.64 = \frac{m'_{sugar}}{m'_{sugar} + 1500} \Rightarrow m_{sugar} = 2666.67 \text{ g}$$

Or

$$C_{water} = \frac{m_{water}}{m_{sugar} + m_{water}}, 0.36 = \frac{1500}{m'_{sugar} + 1500} \Rightarrow m_{sugar} = 2666.67 \text{ g}$$

So, the mass of sugar that will come out of solution upon cooling to 20 °C,  $m''_{sugar}$ :

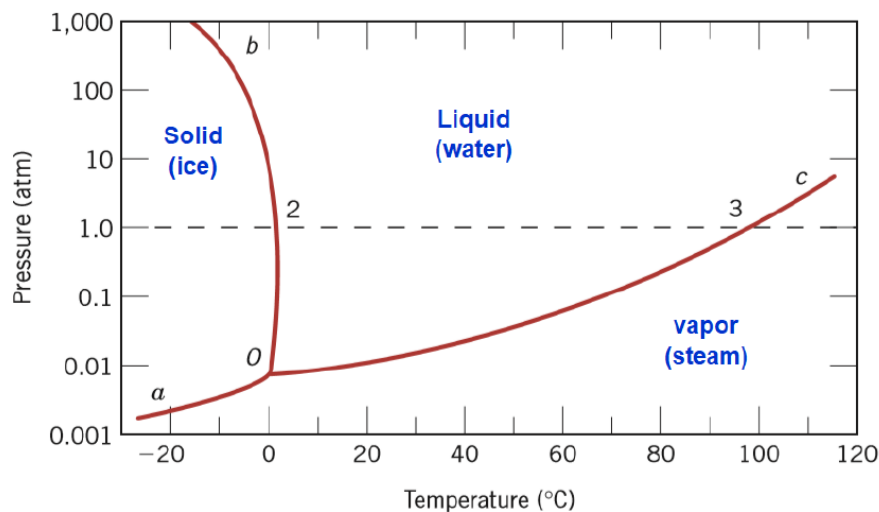
$$m''_{sugar} = m_{sugar} - m'_{sugar} = 5021.74 \text{ g} - 2666.67 \text{ g}$$

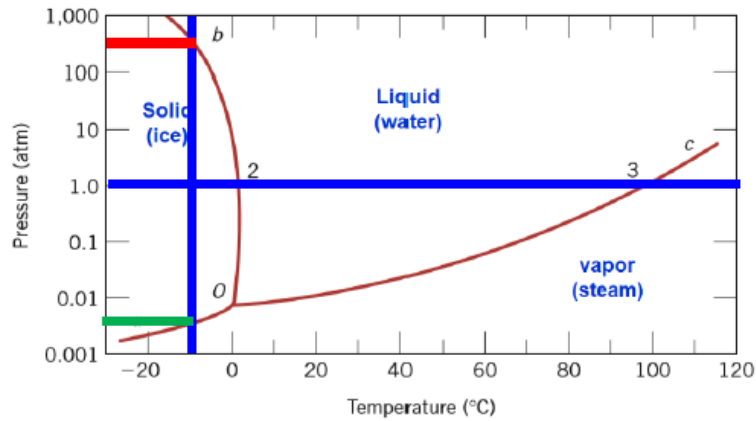
$$\Rightarrow m''_{sugar} = 2355.07 \text{ g}$$

2. Consider a specimen of ice that is at -10 °C and 1 atm pressure. Using the figure of the pressure– temperature phase diagram for H<sub>2</sub>O below, determine the pressure to which the specimen must be raised or lowered to cause it

a. to melt

b. to sublime.





Solution:

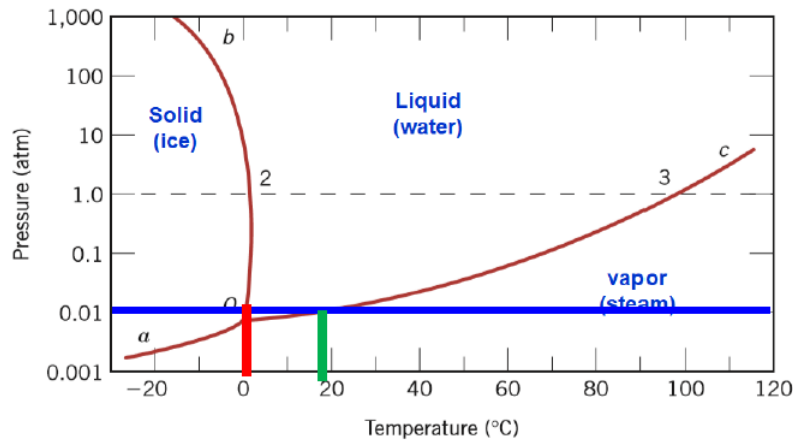
A specimen of ice that is at  $-10^{\circ}\text{C}$  and 1 atm pressure (blue lines).

- a) The pressure at which the specimen melt (from solid phase to liquid phase): is determined by intersection of the blue line with line b (red line): about 570 atm.

terials 2

- b) The pressure at which the specimen sublime (from solid phase to vapor phase): is determined by intersection of the blue line with line a (green line): about 0.005 atm.

3. Using the figure in the previous exercise ( Number 2 ) at a pressure of 0.01 atm, determine
  - a. the melting temperature for ice
  - b. the boiling temperature for water.



Solution:

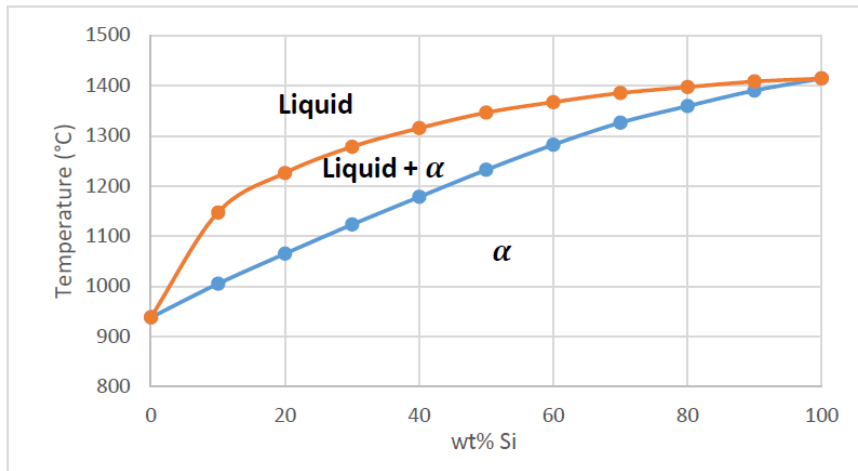
First we draw a blue line at 0.01 atm.

- a) The melting (from solid to liquid phase) temperature for ice at 0.01 atm is the temperature at which the blue line intersects with the red line b (vertical red line): 0.5 °C.
- b) The boiling (from liquid to vapor phase) temperature for water at 0.01 atm is the temperature at which the blue line intersects with the red line c (vertical green line): 17 °C.

4. In the table below, the solidus and liquidus temperatures for the germanium-silicon system are given. Construct the phase diagram for this system and label each region.

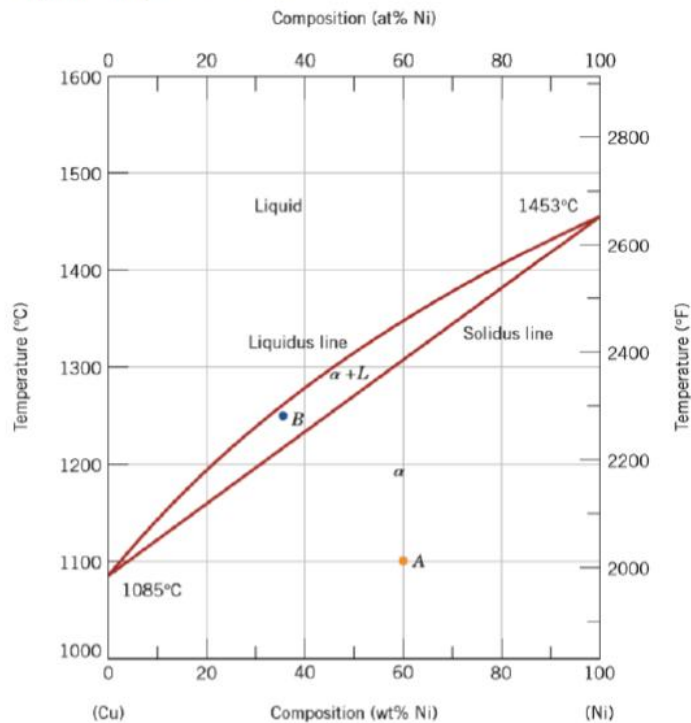
| Composition (wt% Si) | Solidus Temperature (°C) | Liquidus Temperature (°C) |
|----------------------|--------------------------|---------------------------|
| 0                    | 938                      | 938                       |
| 10                   | 1005                     | 1147                      |
| 20                   | 1065                     | 1226                      |
| 30                   | 1123                     | 1278                      |
| 40                   | 1178                     | 1315                      |
| 50                   | 1232                     | 1346                      |
| 60                   | 1282                     | 1367                      |
| 70                   | 1326                     | 1385                      |
| 80                   | 1359                     | 1397                      |
| 90                   | 1390                     | 1408                      |
| 100                  | 1414                     | 1414                      |

Solution:



5. Using the phase diagram below

- Determine the phase(s) present for the Nickel-Copper alloy having 8.2 moles of nickel and 4.3 moles of copper at 1250°C (2280°F). Knowing that the molar mass of nickel is 58.6934 amu and that of copper is 63.5464 amu.
- Compute the relative amount in terms of mass fraction of the phase(s) present for the Nickel-Copper alloy at 1250°C.



Solution:

a) 1 amu = 1g/mol

The masses of nickel and copper are:

$$m_{Ni} = 58.6934 g/mol \times 8.2 moles = 481.28 g$$

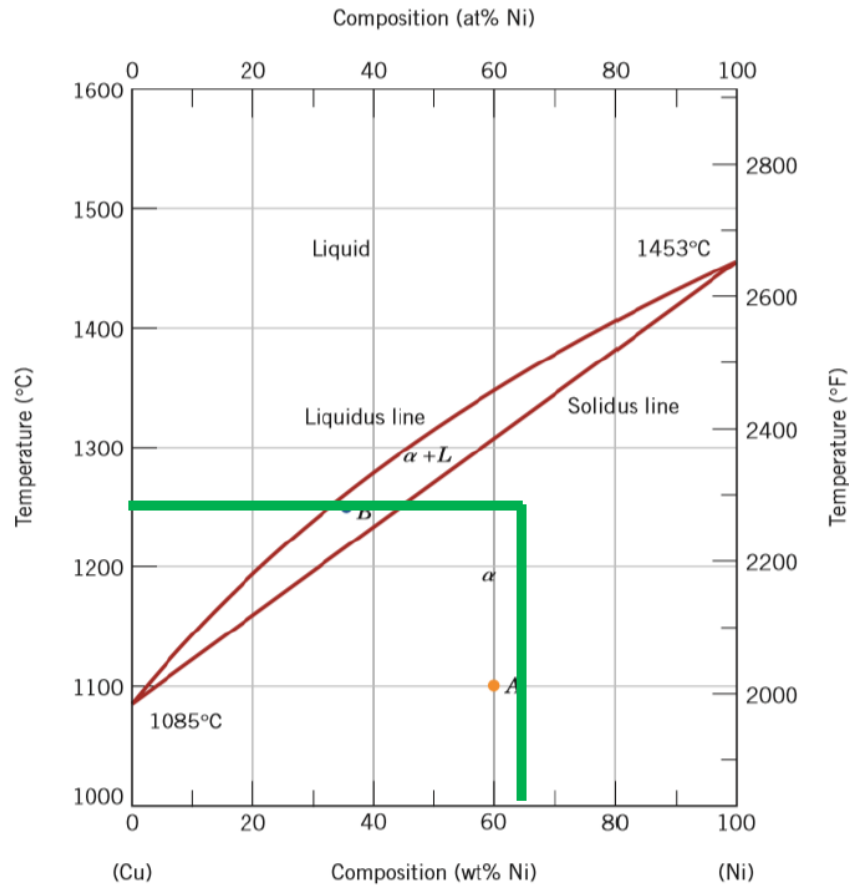
$$m_{Cu} = 63.5464 g/mol \times 4.3 moles = 273.25 g$$

The % compositions of nickel and copper are:

$$C_{Ni} = \frac{m_{Ni}}{m_{Ni} + m_{Cu}} = \frac{481.28 g}{481.28 g + 273.25 g} = 0.64 = 64\%wt Ni$$

$$C_{Cu} = \frac{m_{Cu}}{m_{Ni} + m_{Cu}} = \frac{273.25 g}{481.28 g + 273.25 g} = 0.36 = 36\%wt Cu$$

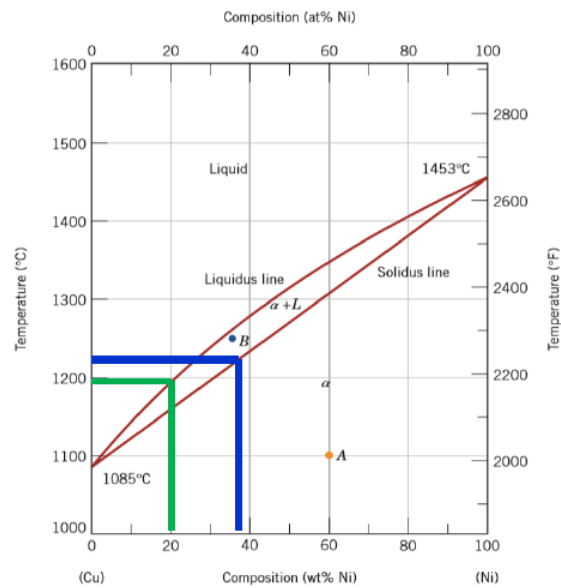
At 1250°C with 64%wt Ni or 36%wt Cu (intersection of the 2 green lines), only  $\alpha$  phase is present.



b) Since only  $\alpha$  phase exist so,

$$W_{\alpha} = 1$$

6. Is it possible to have a copper–nickel alloy that, at equilibrium, consists of a liquid phase of composition 20 wt% Ni–80 wt% Cu and also an  $\alpha$  phase of composition 37 wt% Ni–63 wt% Cu? If so, what will be the approximate temperature of the alloy? If this is not possible, explain why.



Using the Cu-Ni phase diagram above, at a liquid phase of composition 20 wt% Ni–80 wt% Cu, the temperature is 1195°C; however, at an  $\alpha$  phase of composition 37 wt% Ni–63 wt% Cu, the corresponding temperature is 1225°C. There is no unique horizontal line (same temperature) for both compositions.

Therefore, it is impossible for a Cu-Ni alloy with a liquid phase of composition 20 wt% Ni–80 wt% Cu and an  $\alpha$  phase of composition 37 wt% Ni–63 wt% Cu to be at equilibrium.



7. Is it possible to have a copper–zinc alloy that, at equilibrium, consists of an  $\epsilon$  phase of composition 80 wt% Zn–20 wt% Cu, and also a liquid phase of composition 95 wt% Zn–5 wt% Cu? If so, what will be the approximate temperature of the alloy? If this is not possible, explain why.

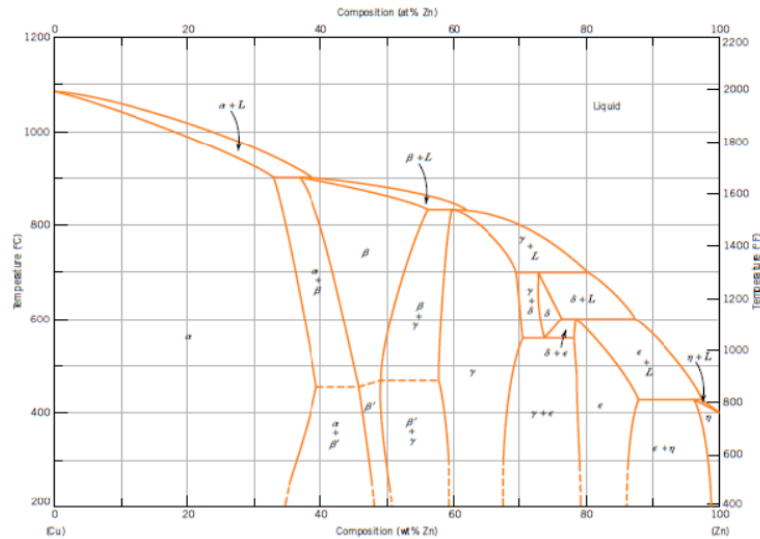


Figure 9.19 The copper–zinc phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 2, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

### Solution

It is *not* possible to have a Cu–Zn alloy, which at equilibrium consists of an  $\epsilon$  phase of composition 80 wt% Zn–20 wt% Cu and also a liquid phase of composition 95 wt% Zn–5 wt% Cu. From Figure 9.19 a single tie line does not exist within the  $\epsilon + L$  region which intersects the phase boundaries at the given compositions. At 80 wt% Zn, the  $\epsilon$ –( $\epsilon + L$ ) phase boundary is at about 575°C, whereas at 95 wt% Zn the ( $\epsilon + L$ )–L phase boundary is at about 490°C.

8. 1.5-kg specimen of a 90 wt% Pb–10 wt% Sn alloy is heated to 250°C (480°F); at this temperature it is entirely a phase solid solution (Figure 9.8). The alloy is to be melted to the extent that 50% of the specimen is liquid, the remainder being the phase. This may be accomplished either by heating the alloy or changing its composition while holding the temperature constant.
- To what temperature must the specimen be heated?
  - How much tin must be added to the 1.5-kg?

### Solution

(a) Probably the easiest way to solve this part of the problem is by trial and error--that is, on the Pb-Sn phase diagram (Figure 9.8), moving vertically at the given composition, through the  $\alpha + L$  region until the tie-line lengths on both sides of the given composition are the same. This occurs at approximately 295°C (560°F).

(b) We can also produce a 50% liquid solution at 250°C, by adding Sn to the alloy. At 250°C and within the  $\alpha + L$  phase region

$$C_{\alpha} = 14 \text{ wt\% Sn-86 wt\% Pb}$$

$$C_L = 34 \text{ wt\% Sn-66 wt\% Pb}$$

Let  $C_0$  be the new alloy composition to give  $W_{\alpha} = W_L = 0.5$ . Then,

$$W_{\alpha} = 0.5 = \frac{C_L - C_0}{C_L - C_{\alpha}} = \frac{34 - C_0}{34 - 14}$$

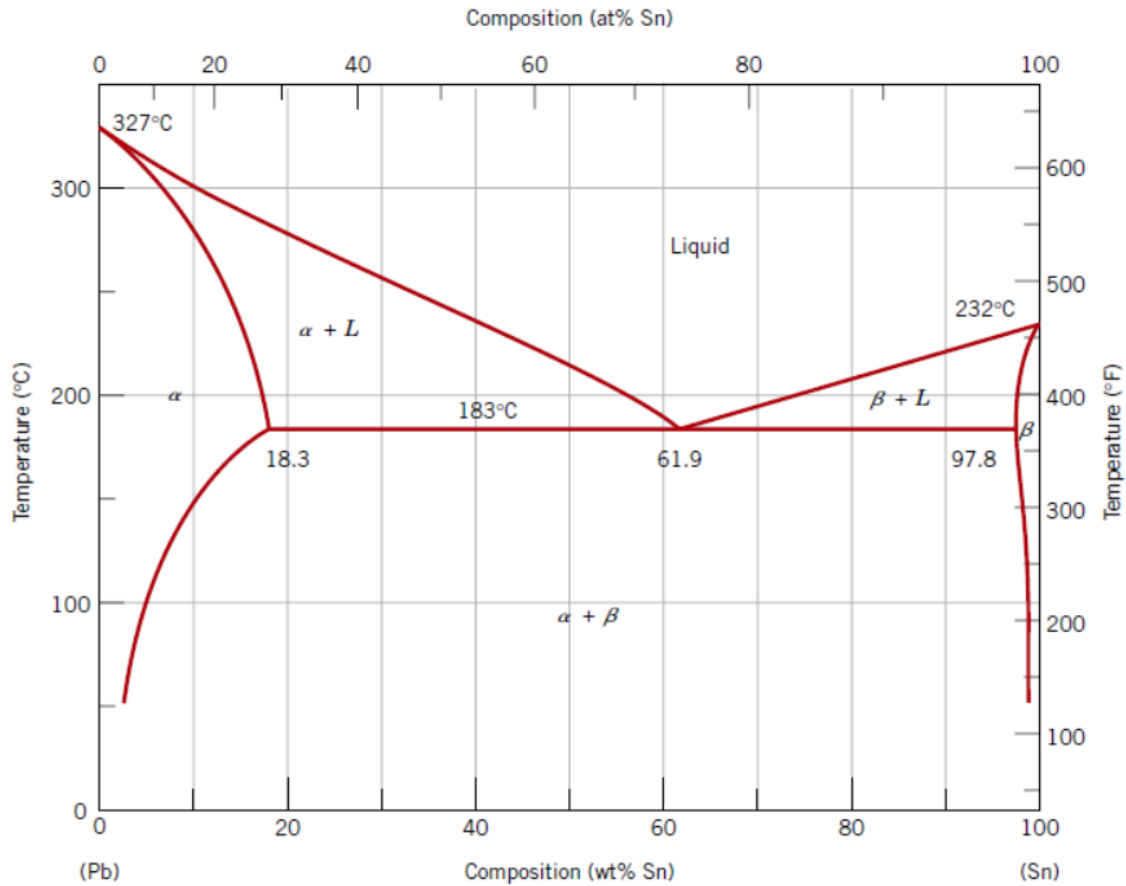
And solving for  $C_0$  gives 24 wt% Sn. Now, let  $m_{\text{Sn}}$  be the mass of Sn added to the alloy to achieve this new composition. The amount of Sn in the original alloy is

$$(0.10)(1.5 \text{ kg}) = 0.15 \text{ kg}$$

[Then, using a modified form of Equation 4.3

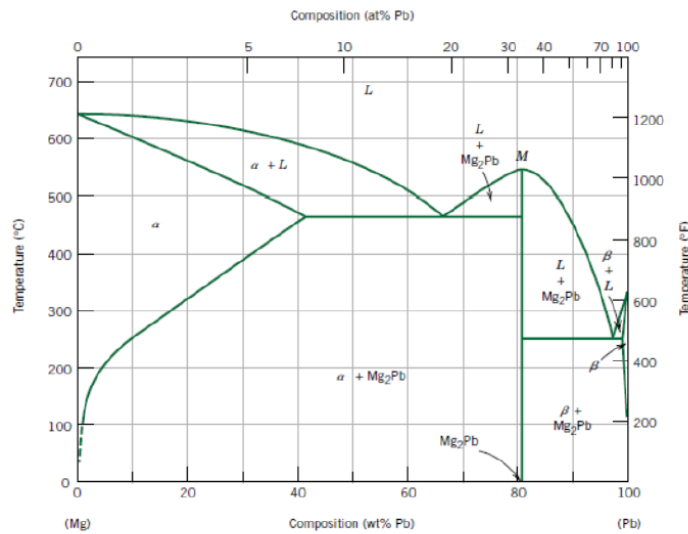
$$\left[ \frac{0.15 \text{ kg} + m_{\text{Sn}}}{1.5 \text{ kg} + m_{\text{Sn}}} \right] \times 100 = 24$$

And, solving for  $m_{\text{Sn}}$  (the mass of tin to be added), yields  $m_{\text{Sn}} = 0.276 \text{ kg}$ .



**Figure 9.8** The lead–tin phase diagram. [Adapted from *Binary Alloy Phase Diagrams*, 2nd edition, Vol. 3, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

9. A magnesium–lead alloy of mass 5.5 kg consists of a solid  $\alpha$  phase that has a composition just slightly below the solubility limit at 200°C (390°F).
- What mass of lead is in the alloy?
  - If the alloy is heated to 350°C (660°F), how much more lead may be dissolved in the  $\alpha$  phase without exceeding the solubility limit of this phase?



**Figure 9.20** The magnesium–lead phase diagram. [Adapted from *Phase Diagrams of Binary Magnesium Alloys*, A. A. Nayeb-Hashemi and J. B. Clark (Editors), 1988. Reprinted by permission of ASM International, Materials Park, OH.]

### Solution

(a) This portion of the problem asks that we calculate, for a Pb–Mg alloy, the mass of lead in 5.5 kg of the solid  $\alpha$  phase at 200°C just below the solubility limit. From Figure 9.20, the solubility limit for the  $\alpha$  phase at 200°C corresponds to the position (composition) of the  $\alpha$ – $\alpha + \text{Mg}_2\text{Pb}$  phase boundary at this temperature, which is about 5 wt% Pb. Therefore, the mass of Pb in the alloy is just  $(0.05)(5.5 \text{ kg}) = 0.28 \text{ kg}$ .

(b) At 350°C, the solubility limit of the  $\alpha$  phase increases to approximately 25 wt% Pb. In order to determine the additional amount of Pb that may be added ( $m_{\text{Pb}}$ ), we utilize a modified form of Equation 4.3 as

$$C_{\text{Pb}} = 25 \text{ wt\%} = \frac{0.28 \text{ kg} + m_{\text{Pb}}}{5.5 \text{ kg} + m_{\text{Pb}}} \times 100$$

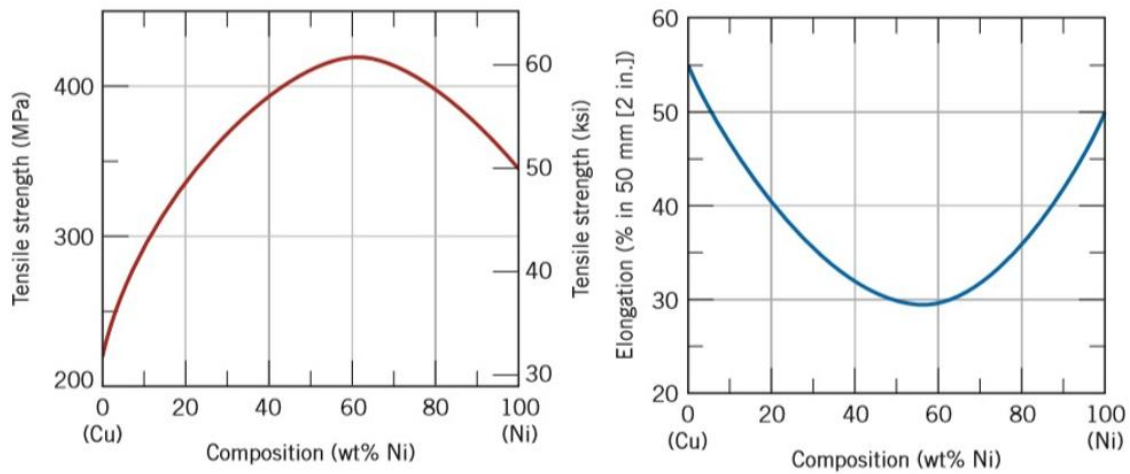
Solving for  $m_{\text{Pb}}$  yields  $m_{\text{Pb}} = 1.46 \text{ kg}$ .

10. It is desirable to produce a copper–nickel alloy that has a minimum noncold-worked tensile strength of 350 MPa (50,750 psi) and a ductility of at least 48%EL. Is such an

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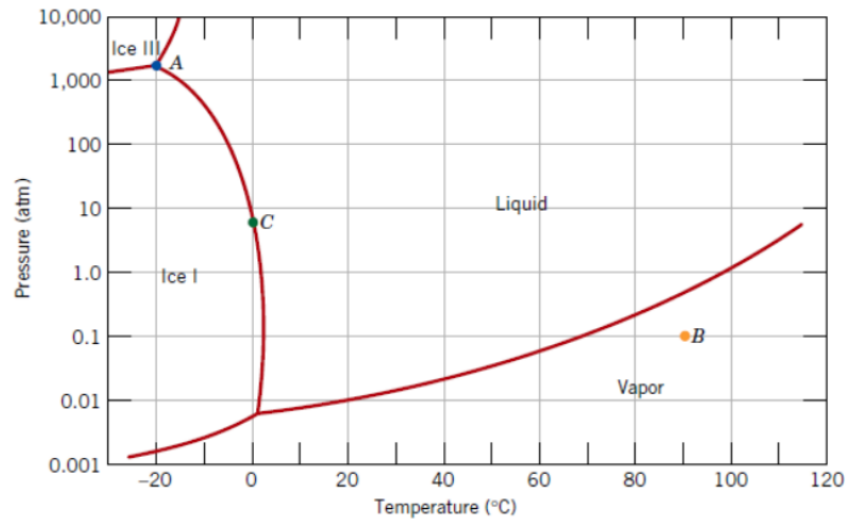
Page 5 Materials 2

alloy possible? If so, what must be its composition? If this is not possible, then explain why.



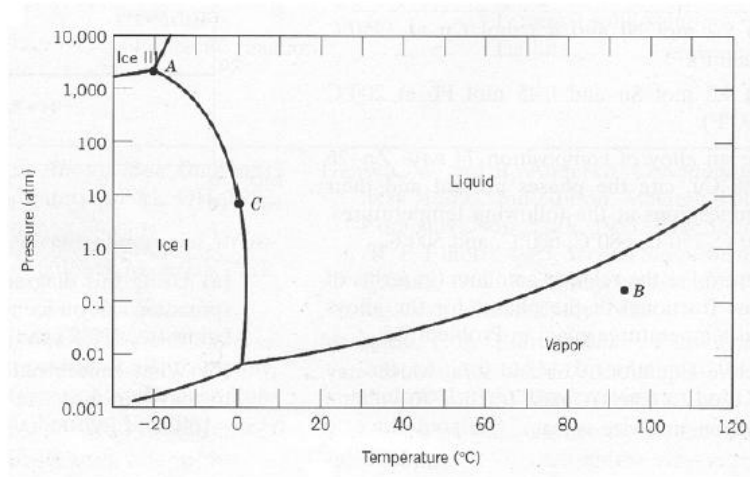
11. Figure 9.38 shows the pressure–temperature phase diagram for  $\text{H}_2\text{O}$ . Apply the Gibbs phase rule at points  $A$ ,  $B$ , and  $C$ ; that is, specify the number of degrees of freedom at each of the points—that is, the number of externally controllable variables that need be specified to completely define the system.

**Figure 9.38**  
Logarithm  
pressure-versus-  
temperature phase  
diagram for  $\text{H}_2\text{O}$ .



### Solution

We are asked to specify the value of  $F$  for Gibbs phase rule at points A, B, and C on the pressure-temperature diagram for  $H_2O$ , Figure 9.38, which is shown below.



Gibbs phase rule in general form is

$$P + F = C + N$$

For this system, the number of components,  $C$ , is 1, whereas  $N$ , the number of noncompositional variables, is 2--viz. temperature and pressure. Thus, the phase rule now becomes

$$P + F = 1 + 2 = 3$$

Or

$$F = 3 - P$$

where  $P$  is the number of phases present at equilibrium.

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At point A, three phases are present (viz. ice I, ice III, and liquid) and  $P = 3$ ; thus, the number of degrees of freedom is zero since

$$F = 3 - P = 3 - 3 = 0$$

Thus, point A is an invariant point (in this case a triple point), and we have no choice in the selection of externally controllable variables in order to define the system.

At point B on the figure, only a single (vapor) phase is present (i.e.,  $P = 1$ ), or

$$F = 3 - P = 3 - 1 = 2$$

which means that specification of both temperature and pressure are necessary to define the system.

And, finally, at point C which is on the phase boundary between liquid and ice I phases, two phases are in equilibrium ( $P = 2$ ); hence

$$F = 3 - P = 3 - 2 = 1$$

Or that we need to specify the value of either temperature or pressure, which determines the value of the other parameter (pressure or temperature).